

Lecture Notes

Course: Complex Analysis

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Topic: Cauchy-Riemann equations

Theorem (Cauchy Riemann). Let Ω be an open subset of $\mathbb{C}$, let $u, v : \Omega \rightarrow \mathbb{R}$ and define the function $f : \Omega \rightarrow \mathbb{C}$ by $f(z) = u(x, y) + iv(x, y)$. If f is complex differentiable at z_0 , then the partial derivatives of u and v all exist, and satisfy the Cauchy Riemann conditions at $z_0 = x_0 + iy_0$, i.e.

$$\partial_x u(x_0, y_0) = \partial_y v(x_0, y_0), \quad \partial_y u(x_0, y_0) = -\partial_x v(x_0, y_0),$$

conversely, if the partial derivatives of u and v all exist, are continuous and satisfy the Cauchy Riemann conditions at (x_0, y_0) , then f is complex differentiable at z_0 .

Proof. $\Rightarrow$] Since f is complex differentiable, the limit

$$\lim_{h \rightarrow 0} \frac{f(z_0 + h) - f(z_0)}{h} = f'(z_0),$$

exists (where $h = h_1 + ih_2$) and is independent of the trajectory, that is, it must yield the same for $h = h_1 \rightarrow 0$ and $h = ih_2 \rightarrow 0$. For $h = h_1$ we get

$$\lim_{h_1 \rightarrow 0} \frac{f(z_0 + h_1) - f(z_0)}{h_1} = \lim_{h_1 \rightarrow 0} \frac{u(x_0 + h_1, y_0) - u(x_0, y_0)}{h_1} + i \lim_{h_1 \rightarrow 0} \frac{v(x_0 + h_1, y_0) - v(x_0, y_0)}{h_1},$$

which we immediately indentify as the partial derivatives of u and v , respectively

$$f'(z_0) = \partial_x u(x_0, y_0) + i \partial_x v(x_0, y_0),$$

and similarly for $h = ih_2$ we get

$$\lim_{h_2 \rightarrow 0} \frac{f(z_0 + ih_2) - f(z_0)}{ih_2} = \lim_{h_2 \rightarrow 0} \frac{u(x_0, y_0 + h_2) - u(x_0, y_0)}{ih_2} + i \lim_{h_2 \rightarrow 0} \frac{v(x_0, y_0 + h_2) - v(x_0, y_0)}{ih_2},$$

$$f'(z_0) = -i \partial_y u(x_0, y_0) + \partial_y v(x_0, y_0)$$

and therefore equating the previous results it is readily seen that we require

$$\partial_x u(x_0, y_0) = \partial_y v(x_0, y_0) \quad \text{and} \quad \partial_x v(x_0, y_0) = -\partial_y u(x_0, y_0).$$

Remark. A function $f : U \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ is said to be differentiable at $\mathbf{x}_0 \in U$ if there exists a vector A in $\mathbb{R}^n$ for which

$$f(\mathbf{x}_0 + \mathbf{h}) - f(\mathbf{x}_0) = A \cdot \mathbf{h} + |\mathbf{h}|E(\mathbf{h})$$

with $\lim_{h \rightarrow 0} E(h) = 0$.

We define the **gradient** of a function $f : U \rightarrow \mathbb{R}$ as

$$\nabla f(\mathbf{x}_0) = \mathbf{e}_i \partial_i f(\mathbf{x}_0).$$

Theorem. If f is differentiable at a point $\mathbf{x}_0$ and let A be a vector such that

$$f(\mathbf{x}_0 + \mathbf{h}) - f(\mathbf{x}) = A \cdot \mathbf{h} + |\mathbf{h}|E(\mathbf{h}).$$

Then all partial derivatives of f at $\mathbf{x}_0$ exists and $A = \nabla f(\mathbf{x}_0)$.

$\Leftarrow$] Let $f(z) = u(x, y) + iv(x, y)$ and suppose that u, v are at both least class C^1 , then definition of differentiability at (x_0, y_0) yields

$$u(x_0 + h_x, y_0 + h_y) - u(x_0, y_0) = \nabla u(\mathbf{x}_0) \cdot \mathbf{h} + |\mathbf{h}|E_u(\mathbf{h})$$

$$v(x_0 + h_x, y_0 + h_y) - v(x_0, y_0) = \nabla v(\mathbf{x}_0) \cdot \mathbf{h} + |\mathbf{h}|E_v(\mathbf{h})$$

with $E_u \rightarrow 0$ and $E_v \rightarrow 0$ as $h \rightarrow 0$. Now, consider $f(z_0 + h) - f(z_0)$ ($h = h_x + ih_y$) which is just

$$\begin{aligned} f(z_0 + h) - f(z_0) &= u(x_0 + h_x, y_0 + h_y) - u(x_0, y_0) + iv(x_0 + h_x, y_0 + h_y) - iv(x_0, y_0) \\ &= \partial_x u(x_0, y_0)h_x + \partial_y u(x_0, y_0)h_y + |h|E_u(h) + i\partial_x v(x_0, y_0)h_x + i\partial_y v(x_0, y_0)h_y + |h|E_v(h) \\ &= \partial_x u(x_0, y_0)(h_x + ih_y) + \partial_x v(x_0, y_0)(ih_x - h_y) + |h|(E_u(h) + E_v(h)) \\ &= [\partial_x u(x_0, y_0) + i\partial_x v(x_0, y_0)]h + |h|(E_u(h) + E_v(h)) \end{aligned}$$

so that

$$\lim_{h \rightarrow 0} \frac{f(z_0 + h) - f(z_0)}{h} = \lim_{h \rightarrow 0} \left[\partial_x u(x_0, y_0) + i\partial_x v(x_0, y_0) + \frac{|h|}{h}(E_u(h) + E_v(h)) \right].$$

We conclude that f is differentiable at z_0 and $f'(z_0) = \partial_x u(x_0, y_0) + i\partial_x v(x_0, y_0)$.